NASCOMSOFT EMBEDDED HUB | DATA DEPARTMENT

Network Analyser: 1-Week MVP Build

Accelerated Implementation Roadmap & Scope Definition

Objective: Transition immediately from product discovery to a compressed 7-day engineering sprint. The goal is to deploy a functional Minimum Viable Product (MVP) of the Network Analyser to a controlled pilot network by Day 7. This timeline requires ruthless prioritization, strict adherence to the locked scope, and immediate technical execution without over-engineering.

The product remains focused on providing authorized Starlink network administrators with actionable visibility into bandwidth consumption, device connections, and network health—without crossing into covert surveillance.

1. Strict Scope Lockdown

To survive a one-week build, the engineering team must hold a hard line on deliverables. Any feature not explicitly listed in the "In Scope" column is deferred to Post-Pilot phases.

In Scope for Week 1 (MVP)	Strictly Out of Scope
INCLUDED **Connected-device discovery and basic IP/MAC mapping**	DEFERRED AI traffic classification & deep packet inspection
INCLUDED **Per-device upload/download consumption trends**	PROHIBITED Content inspection, reading messages, or file tracking
INCLUDED **Network health metrics (latency, speed tests, uptime)**	DEFERRED Automated blocking or bandwidth throttling
INCLUDED **Simple configurable alerts (unknown device, high usage)**	DEFERRED Multi-site enterprise management & billing integration

2. Privacy & Security Directives

Mandatory Safeguards for MVP

- **No Payload Inspection:** The analyzer must operate purely on headers and metadata. Never inspect payload contents.
- **Data Minimization:** Store only the telemetry required to render the 7-day consumption dashboards. Drop granular flow data after 24 hours.
- **Notice Requirement:** The pilot UI must include a prominent administrator acknowledgment that the tool is being used on an authorized, owned network.

3. The 7-Day Implementation Sprint

The daily cadence shifts from research to rapid build. Each day focuses on a distinct architectural pillar, leading to a functional pilot deployment by Day 7.

Day 1: Architecture Lock & Foundation

Activities: Finalize the deployment option (router integration vs. hybrid). Lock the primary database schema and establish the local development environments and code repositories.

Expected Outcome: A documented and locked database schema, provisioned staging cloud environments, and a running local development setup ready for feature branches.

Day 2: Core Ingestion Engine

Activities: Build the backend data ingestion pipelines. Implement connected-device discovery protocols, device naming algorithms, and the initial IP/MAC address mapping logic.

Expected Outcome: The backend system successfully discovers simulated or local connected devices and continuously logs their identities into the database without duplication.

Day 3: Telemetry & Health Tracking

Activities: Implement tracking mechanisms for per-device upload and download bandwidth usage. Integrate API endpoints or local listeners to capture basic health metrics like latency, speed test results, and connection uptime.

Expected Outcome: Continuous, accurate capture of bandwidth and health metrics stored efficiently in a time-series format for rapid frontend querying.

Day 4: Admin Dashboards & UI

Activities: Develop the frontend network-wide usage dashboard. Build the peak-usage timeline visualization and implement basic role-based access control (RBAC).

Expected Outcome: A functional, authenticated web interface where administrators can view live and historical bandwidth consumption charts, alongside an accurate device list.

Day 5: Alerts & Export Reporting

Activities: Configure logic triggers for unknown-device detection and high-usage thresholds. Build the report generation engine to export daily and weekly consumption into manager-friendly CSV/PDF formats.

Expected Outcome: Automated UI/email alerts successfully fire upon new device connections, and administrators can instantly download formatted usage reports for review.

Day 6: Privacy Hardening & QA

Activities: Audit the entire codebase against the established privacy principles. Conduct end-to-end testing to verify data accuracy, ensure encrypted traffic remains untouched, and patch any critical bugs.

Expected Outcome: A stable, fully tested MVP verified to drop payload data, complete with a visible, mandatory privacy notice implemented for first-time administrators.

Day 7: Pilot Launch & Monitoring

Activities: Deploy the MVP to the designated internal or partner pilot network. Monitor ingestion stability, system memory usage, and provide initial access credentials to pilot users.

Expected Outcome: A live, active deployment on the target network with telemetry flowing correctly, causing zero disruption or noticeable latency to the host Starlink connection.

4. Pilot Success Criteria

Day 7+ Evaluation Metrics

To determine if the MVP is successful enough to graduate from the pilot phase, track the following immediate metrics upon launch:

- **Accuracy:** Device identification and data-consumption tracking match physical reality with >90% accuracy.
- **Performance:** Data ingestion does not introduce noticeable latency or bottlenecking to the host Starlink connection.
- **Actionability:** The administrator successfully uses the dashboard to identify the top 3 bandwidth consumers within the first 48 hours.

Post-Launch Action

On Day 8, convene the Data Lead and Engineering team for a retrospective. Review the real-world data populated in the dashboard and prioritize the Day 8-14 backlog (critical bug fixes vs. minor usability enhancements).